

Coulomb's Law: Force Between Charges

SimuPhysics Coulomb's Law Sandbox · SimuPhysics

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://simuphysics.com/view/electricity-and-magnetism-coulomb-s-law-sandbox.html>
Curriculum links: electric fields · electricity

Learning intentions

- I can predict the direction of the electric force between two charges based on their signs.
- I can describe how electric force changes with distance and charge magnitude.
- I can use Coulomb's Law to calculate the force between two point charges.

Getting started

Open the SimuPhysics Coulomb's Law Sandbox. Place two charged particles in the workspace, starting with two positive charges of equal magnitude. Use the distance and charge controls to change their separation and charge values, checking the force readout after each prediction.

Tasks

1. With two positive charges placed a moderate distance apart, predict whether they will attract or repel, then check the simulation. Record the direction of the force.

2. Change one charge from positive to negative, keeping the same distance and magnitude. Predict and then record the new force direction.

3. Keeping both charge magnitudes constant, move the charges to five different separations. Record the distance and force magnitude at each.

Distance between charges	Force magnitude reading

4. Using your table, describe how force changes as distance increases. Does doubling the distance halve the force, or change it by a different factor?

5. Choose two of your trials from Task 3 where the distance roughly doubles. Calculate the ratio of the two force readings. Does this match what Coulomb's Law ($F \propto 1/r^2$) predicts?

Working:

6. Now keep distance constant and vary the magnitude of one charge across four values. Record the charge values and resulting force.

Charge on particle 1	Charge on particle 2	Force magnitude reading

7. Based on Task 6, describe the relationship between charge magnitude and force.

8. Using Coulomb's Law, $F = k \cdot q_1 \cdot q_2 / r^2$, calculate the expected force between two charges of $+2 \mu\text{C}$ and $-3 \mu\text{C}$ separated by 0.5 m ($k \approx 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$). Compare the order of magnitude of your answer with a similar setup in the simulation.

Working:

Extension (optional)

Explain why the electric force between two protons in an atomic nucleus is enormous at such tiny distances, yet everyday charged objects (like a rubbed balloon) only exert a noticeable force over centimetres. Use the distance relationship you found in this worksheet in your explanation.
